Unpacking the Demand for Sustainable Equity Investing

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Background: Sustainable (ESG) Investing

1. Demand for ESG investing: Grown rapidly over past decade. By how much exactly is unclear...

2. Asset Price Effects: Does ESG demand increase valuation of sustainable firms and lower their expected returns (cost of capital)?
    - Standard Approach: Estimate *expected returns* of ESG portfolios

      Bolton and Kacpercyk (2021), Edmans et al (2012), Pastor, Stambaugh, Taylor (2021), Pedersen et al (2020)...
    - New Wave: Estimate *price impact* due to ESG investors

      Berk and Binsbergen (2021), Koijen, Richmond, Yogo, (2023)...

3. Real Effects: Do firms improve environmental performance (e.g. more sustainable investments via lower cost of capital)?

   Hartzmark and Shu (2023), Harris (2023)

The paper does **a lot**!

1. Identify investor demand for 3 different sustainability characteristics
   $\rightarrow$ Seperating demand shifter from slope via demand system
2. Disaggregate ESG demand: Who is behind the demand shift?
   $\rightarrow$ Surprising: The most elastic investors (the arbitrageurs)!
3. Estimate asset price effects of sustainability demand
   $\rightarrow$ 1 StDev $\downarrow$ in Scope1 CO2 emissions leads to 8.4% $\uparrow$ market value!
4. Estimate real effects of sustainability demand
   $\rightarrow$ Sustainability demand predicts environmental improvement
5. Counterfactuals: What if ESG scores reflected Green Patents
   $\rightarrow$ Market value of top green innovators 4.7% $\uparrow$

Comments

Very nice paper!

- Well-executed and well-written
- Provides a detailed picture of investors' demand for sustainability and its (pricing/real) implications
- It combines many current questions: Stimulating to read

This Discussion

- Small note on identification
- Why don't you split it in (at least) two papers and narrow the focus?
    1. Unpacking demand for ESG: How much, for what, who and why?
    2. The impact of demand for ESG: Asset pricing and real effects

The Setting

- Notation: Stock n, quarter t, investor i
- Regression of market equity $ME_t(n)$ onto ESG score $s_t(n)$, controlling for book equity and other fundamentals

$$ME_t(n) = \beta_{1,t} s_t(n) + Controls + \epsilon_t(n)$$

$\rightarrow \beta_{1,t}$ high and increasing over time. What does this mean?

1. Sustainable companies have increasingly higher valuations.
2. But can't say much more: From CFO's perspective, increasing your $s_t(n)$ does not necessarily lead to higher valuation

Simplified Linear Demand System

- Instead of aggregate regression, run investor-specific regressions

$$w_{it}(n) = \beta_{0,it} ME_t(n) + \beta_{1,it} s_t(n) + Controls + \epsilon_{it}(n)$$

  using some valid instrument for endogenous prices $ME_t(n)$.

- Market clearing $\sum_i w_{it}(n) A_{it} = ME_t(n)$ implies

$$ME_t(n) = \underbrace{\boxed{\frac{\sum_i A_{it}\beta_{1,it}}{1 - \sum_i A_{it}\beta_{0,it}}}}_{\text{Investor Pressure } \beta_{1,t}} s_t(n) + Controls + \epsilon_t(n)$$

- Demand system allows decomposing coefficient $\beta_{1,t}$ (e.g. how much is it driven by ESG demand of investor i?)
- ... But estimating investor-specific $\beta_{1,it}$ instead of aggregate $\beta_{1,t}$ directly still does not solve the identification problem!

A note on Identification

A simple example

- Investor i has demand for AI firms (which have high ESG scores), resulting in positive $\beta_{1,it}$ and positive investor pressure $\beta_{1,t}$ for ESG
- Counterfactual experiment:
  1. Firm n hires Yoga coach and improves ESG score: $\Delta s_t(n)$
  2. Counterfactual change in valuation: $\Delta ME_t(n) = \Delta s_t(n)\beta_{1,t}$
  3. But the firm didn't become an AI firm: overestimated $\Delta ME_t(n)$

An alternative way of expressing this issue

- An instrument for the $ME_t(n)$ in the demand estimation allows disentangling demand *slope* and *shifter*

$$w_{it}(n) = \overbrace{\beta_{0,it} ME_t(n)}^{\textit{Slope}} + \overbrace{\beta_{1,it} s_t(n) + Controls + \epsilon_{it}(n)}^{\textit{Shifter}}$$

- But this paper is about decomposing the shifter into ESG demand and non-ESG demand!

Paper 1: Unpacking Demand for Sustainability

1. How much? $\sim$
    - Challenge: How can there be an *aggregate* demand shift (i.e. tilt) towards ESG? For every buyer there is a seller.
    - Answer: Demand shift by one investor offset by movement along the curve of another. ✓
    - Convert aggregate demand coefficient into $!
    - Out of curiosity: what's doing the work? Coefficient constraint or instrument?

2. For what? ✓
    - There is aggregate institutional demand for sustainability *beyond* readily available scores

Paper 1: Unpacking Investor Demand

3. Who? ✓
   - The most elastic investors! Wow!
   - But tell us more!

4. Why?
   - Pecuniary versus non-pecuniary preferences (value versus values)
   - ESG demand comes from arbitrageurs $\rightarrow$ Pecuniary?

Paper 2: Asset Pricing and Real Effects

- ESG demand $\rightarrow$ Asset Prices $\rightarrow$ Environmental Performance
- This is a **super important** question: Probably difficult to answer in section 5.2
- But model-based investor-pressure variable is a cool starting point!

Other comments

… some other comments and thoughts

Investor pressure in the Cross-Section

$$\text{Investor Pressure}_t(n) \approx \underbrace{\text{ESG Demand}_t(n)}_{\log \sum_i A_{it}\beta_{it}} \times \underbrace{\text{Price Impact}_t(n)}_{\log(1-\sum_i A_{it}\beta_{0,it})}$$

- CFO's incentive to improve sustainability driven by two things:
  1. How much do my investors like sustainability?
  2. How many active hedge funds are in my investor base?
- You show that ESG investors are the most elastic: So effects partly offset each other in the cross-section!
- But are CFOs aware of that? Simple regression

$$\Delta s_{t+1}(n) = \alpha_1 \text{ESG Demand}_t(n) + \alpha_2 \text{Price Impact}_t(n) \times \mathbf{1}_{\text{ESG Demand}>0} + \ldots$$

- "Theory" implies α_1 and α_2 positive

Investor pressure and environmental performance

Why would investor pressure $\frac{\sum_i A_{it}\beta_{1,it}}{1-\sum_i A_{it}\beta_{0,it}}$ forecast environmental performance?

- Implicitly assumes that firms are not at the optimum: Increasing sustainability would increase shareholder value
- But increasing sustainability likely comes at a cost (e.g. expensive green energy and therefore lower profitability)
- Big step: Endogenizing corporate decision-making in demand-based asset pricing models.
- Say you find big effects: Would this violate the initial assumption that all characteristics except price are driven by an exogenous endowment process?

Potential instruments for ESG scores:

1. Many scores based on relative rankings. So when *m* firm improves Co2 emissions, *n*'s ESG score may change while leaving fundamentals unchanged
2. Portfolios of labelled ESG funds: Difference between elastic (pecuniary) and inelastic (non-pecuniary)
3. Exogenous ESG scores around arbitrary cutoffs (akin to Morningstar globes)

Conclusion

Very rich paper

- Carefully executed, well written
- Contributions to many areas of sustainable finance
    1. Understanding sustainability demand: Source, magnitude, direction
    2. Asset pricing implications
    3. Real effects
- Consider spliting in two papers and narrow the focus